

Commerce XI Maths App

Theory (Pages 1–9)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 1 - Sets and Relations | Total questions: 35

Question 1 - Theory Q1

Which collection is a set?

Options

- A. Days in a week
- B. Happy people in a town
- C. Clever students in a class
- D. Successful people in a city

Correct answer: Days in a week

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Days in a week. The other choices omit a required element, include an invalid element, or use a different operation.

Question 2 - Theory Q2

A set is a collection of objects that are:

Options

- A. Well-defined
- B. Similar
- C. Countable
- D. Written in order

Correct answer: Well-defined

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Well-defined. The other choices omit a required element, include an invalid element, or use a different operation.

Question 3 - Theory Q3

If x is an element of A , which notation is correct?

Options

- A. $x \in A$
- B. $x \subseteq A$
- C. $A \in x$
- D. $x \notin A$

Correct answer: x A

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $x \in A$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 4 - Theory Q4

In the Roster method, elements are:

Options

- A. Listed inside braces and separated by commas
- B. Shown only in a diagram
- C. Described by a property
- D. Listed without brackets

Correct answer: Listed inside braces and separated by commas

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Listed inside braces and separated by commas. The other choices omit a required element, include an invalid element, or use a different operation.

Question 5 - Theory Q5

What is the set of letters in ABBA in Roster form?

Options

- A. {A, B}
- B. {A, B, B, A}
- C. {A, A, B, B}
- D. {AB, BA}

Correct answer: {A, B}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {A, B}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 6 - Theory Q6

Does changing the order of elements change a set?

Options

- A. No

- B. Yes
- C. Only for finite sets
- D. Only for number sets

Correct answer: No

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives No. The other choices omit a required element, include an invalid element, or use a different operation.

Question 7 - Theory Q7

The Set-Builder method describes elements by:

Options

- A. A property that determines them
- B. Their order
- C. A diagram
- D. Their number alone

Correct answer: A property that determines them

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives A property that determines them. The other choices omit a required element, include an invalid element, or use a different operation.

Question 8 - Theory Q8

A Venn diagram is a:

Options

- A. Pictorial representation of a set
- B. List of every element
- C. Formula for cardinality
- D. Type of interval

Correct answer: Pictorial representation of a set

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Pictorial representation of a set. The other choices omit a required element, include an invalid element, or use a different operation.

Question 9 - Theory Q9

Which interval excludes both a and b?

Options

- A. (a, b)
- B. $[a, b]$
- C. $[a, b)$
- D. $(a, b]$

Correct answer: (a, b)

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives (a, b) . The other choices omit a required element, include an invalid element, or use a different operation.

Question 10 - Theory Q10

Which interval includes both a and b?

Options

- A. $[a, b]$
- B. (a, b)
- C. $[a, b)$
- D. $(a, b]$

Correct answer: $[a, b]$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $[a, b]$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 11 - Theory Q11

Which interval includes a but excludes b?

Options

- A. $[a, b)$
- B. (a, b)
- C. $(a, b]$
- D. $[a, b]$

Correct answer: $[a, b)$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives [a, b]. The other choices omit a required element, include an invalid element, or use a different operation.

Question 12 - Theory Q12

Which interval excludes a but includes b?

Options

- A. $(a, b]$
- B. $[a, b)$
- C. $[a, b]$
- D. (a, b)

Correct answer: $(a, b]$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives [a, b]. The other choices omit a required element, include an invalid element, or use a different operation.

Question 13 - Theory Q13

$n(A)$ means the number of:

Options

- A. Distinct elements in A
- B. Sets equal to A
- C. Subsets of A
- D. Elements outside A

Correct answer: Distinct elements in A

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Distinct elements in A. The other choices omit a required element, include an invalid element, or use a different operation.

Question 14 - Theory Q14

A set with no element is called a:

Options

- A. Empty set
- B. Singleton set
- C. Universal set

D. Power set

Correct answer: Empty set

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Empty set. The other choices omit a required element, include an invalid element, or use a different operation.

Question 15 - Theory Q15

A set containing exactly one element is a:

Options

A. Singleton set

B. Finite set only

C. Universal set

D. Proper subset

Correct answer: Singleton set

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Singleton set. The other choices omit a required element, include an invalid element, or use a different operation.

Question 16 - Theory Q16

Is the empty set finite?

Options

A. Yes

B. No

C. Only when the universal set is finite

D. Only when it is written as $\{ \}$

Correct answer: Yes

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Yes. The other choices omit a required element, include an invalid element, or use a different operation.

Question 17 - Theory Q17

Which is an infinite set?

Options

- A. Natural numbers
- B. Vowels in English
- C. Days in a week
- D. Letters in "beautiful"

Correct answer: Natural numbers

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Natural numbers. The other choices omit a required element, include an invalid element, or use a different operation.

Question 18 - Theory Q18

Two sets are equal when they have:

Options

- A. The same elements
- B. The same number of elements
- C. The same order of elements
- D. The same name

Correct answer: The same elements

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives The same elements. The other choices omit a required element, include an invalid element, or use a different operation.

Question 19 - Theory Q19

Two finite sets are equivalent when they have:

Options

- A. The same number of elements
- B. Exactly the same elements
- C. No common elements
- D. The same power set

Correct answer: The same number of elements

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives The same number of elements. The other choices omit a required element, include an invalid element, or use a different operation.

Question 20 - Theory Q20

$A \subseteq B$ means:

Options

- A. Every element of A is in B
- B. Every element of B is in A
- C. A and B have equal sizes
- D. A is empty

Correct answer: Every element of A is in B

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Every element of A is in B. The other choices omit a required element, include an invalid element, or use a different operation.

Question 21 - Theory Q21

If A is a proper subset of B, then:

Options

- A. Every element of A is in B, and B has at least one element not in A
- B. $A = B$
- C. A has more elements than B
- D. A and B are disjoint

Correct answer: Every element of A is in B, and B has at least one element not in A

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Every element of A is in B, and B has at least one element not in A. The other choices omit a required element, include an invalid element, or use a different operation.

Question 22 - Theory Q22

A universal set for a discussion contains:

Options

- A. Every set under consideration as a subset

- B. Only natural numbers
- C. Only one element
- D. No elements

Correct answer: Every set under consideration as a subset

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Every set under consideration as a subset. The other choices omit a required element, include an invalid element, or use a different operation.

Question 23 - Theory Q23

The power set $P(A)$ contains:

Options

- A. All subsets of A
- B. Only the elements of A
- C. Only proper subsets of A
- D. Only the empty set

Correct answer: All subsets of A

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives All subsets of A. The other choices omit a required element, include an invalid element, or use a different operation.

Question 24 - Theory Q24

Relative to universal set U, A' contains:

Options

- A. Elements in U that are not in A
- B. All elements of A
- C. Elements common to A and U
- D. Elements outside U

Correct answer: Elements in U that are not in A

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Elements in U that are not in A. The other choices omit a required element, include an invalid element, or use a different operation.

Question 25 - Theory Q25

Which identity is correct?

Options

A. $(A')' = A$

B. $(A')' = \emptyset$

C. $(A')' = U$

D. $(A')' = P(A)$

Correct answer: $(A')' = A$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $(A')' = A$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 26 - Theory Q26

$A \cup B$ contains elements in:

Options

A. A or B, including those in both

B. Both A and B only

C. A but not B

D. Neither set

Correct answer: A or B, including those in both

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives A or B, including those in both. The other choices omit a required element, include an invalid element, or use a different operation.

Question 27 - Theory Q27

$A \cap B$ contains elements in:

Options

A. Both A and B

B. A or B

C. A but not B

D. Neither set

Correct answer: Both A and B

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Both A and B. The other choices omit a required element, include an invalid element, or use a different operation.

Question 28 - Theory Q28

If $A \cap B = \emptyset$, the sets are:

Options

- A. Disjoint
- B. Equal
- C. Equivalent
- D. Universal

Correct answer: Disjoint

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives Disjoint. The other choices omit a required element, include an invalid element, or use a different operation.

Question 29 - Theory Q29

If $A = \{a, b, a, c\}$, enter $n(A)$.

Correct answer: 3

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 3. The other choices omit a required element, include an invalid element, or use a different operation.

Question 30 - Theory Q30

If $A = \{p, q, r, s\}$, enter $n(P(A))$.

Correct answer: 16

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 16. The other choices omit a required element, include an invalid element, or use a different operation.

Question 31 - Theory Q31

If $n(A) = 12$, $n(B) = 9$, and $n(A \cap B) = 4$, enter $n(A \cup B)$.

Correct answer: 17

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 17. The other choices omit a required element, include an invalid element, or use a different operation.

Question 32 - Theory Q32

If $n(U) = 25$ and $n(A) = 8$, enter $n(A')$.

Correct answer: 17

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 17. The other choices omit a required element, include an invalid element, or use a different operation.

Question 33 - Theory Q33

If $n(A \cup B) = 18$, $n(A) = 10$, and $n(B) = 12$, enter $n(A \cap B)$.

Correct answer: 4

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 4. The other choices omit a required element, include an invalid element, or use a different operation.

Question 34 - Theory Q34

If $n(A) = 10$ and $n(A \cap B) = 3$, enter $n(A - B)$.

Correct answer: 7

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 7. The other choices omit a required element, include an invalid element, or use a different operation.

Question 35 - Theory Q35

If $n(A) = 12$, $n(B) = 10$, $n(C) = 8$, $n(A \cap B) = 5$, $n(B \cap C) = 4$, $n(A \cap C) = 3$, and $n(A \cap B \cap C) = 2$, enter $n(A \cup B \cup C)$.

Correct answer: 20

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 20. The other choices omit a required element, include an invalid element, or use a different operation.